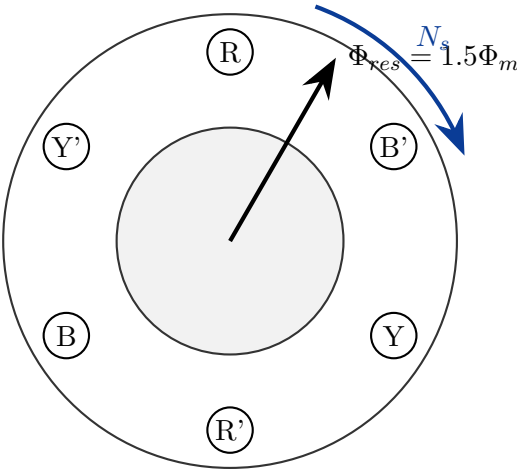


PHASE 1: THE MAGIC OF THE RMF

The Rotating Magnetic Field

A Frame-by-Frame Visual and Mathematical Masterclass

How to create a spinning tornado of magnetic force using absolutely no moving mechanical parts.



Contents

Introduction: The Greatest Trick in Engineering	2
Block 1: The Setup (Space and Time)	2
Block 2: The Vector Rules	3
Block 3: Frame-by-Frame Animation (The Textual Video)	4
Block 4: The Rigorous Mathematical Proof	8
Block 5: Controlling the Beast (Direction and Speed)	10
Phase 1 Completion Checklist	11

Introduction: The Greatest Trick in Engineering

In Phase 0, we learned that a moving magnetic field pushes against a conductor to create voltage (Faraday), and a short-circuited conductor will fight that motion by trying to drag itself along with the field (Lenz).

If we want a heavy steel rotor to spin continuously, we need a magnetic field that spins continuously around it in a circle.

The obvious (but terrible) solution: Take a giant physical horseshoe magnet, mount it on a shaft, and use a separate diesel engine to spin that magnet around the rotor. The rotor would follow it. But this defeats the entire purpose of an electric motor! We want to convert electricity to motion, not motion to motion.

The genius solution (Nikola Tesla's breakthrough): What if we bolt three separate electromagnets to a stationary iron frame (the stator)? The electromagnets cannot physically move. But, by precisely timing *when* we pump electricity into each one, we can create an invisible magnetic wave that sweeps around the circle.

This is the **Rotating Magnetic Field (RMF)**. It is the fundamental operating principle of every three-phase induction and synchronous machine on Earth.

The Neurological Link (Why does this matter?)

Imagine three friends standing in a circle around a heavy pendulum. Friend 1 pushes it toward Friend 2. Friend 2 immediately pushes it toward Friend 3. Friend 3 pushes it back toward Friend 1.

None of the friends are running in a circle. They are standing perfectly still. But because they timed their pushes sequentially, the pendulum swings in a perfect, continuous circle. The friends are the stator coils. Their pushes are the 3-phase AC currents. The swinging pendulum is the Rotating Magnetic Field.

Block 1: The Setup (Space and Time)

To create an RMF, we need two ingredients perfectly synced: a spatial displacement and a temporal (time) displacement.

Ingredient 1: Space (The Stator Construction)

We take three identical sets of copper coils. We embed them into the slots of a circular stationary iron core (the stator).

We do not place them randomly. We space their magnetic axes exactly 120° **apart in space**.

- **Phase R coil axis:** 0° (Reference)
- **Phase Y coil axis:** 120° (Counter-clockwise from R)
- **Phase B coil axis:** 240° (Counter-clockwise from R)

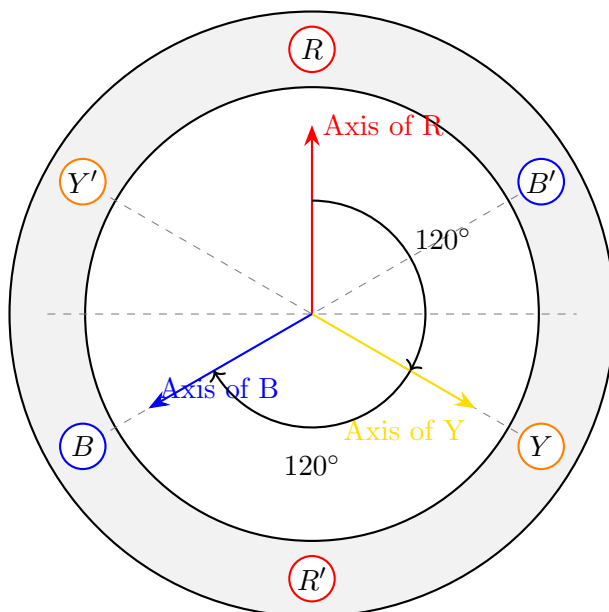


Figure 1: Spatial displacement of the three phase windings on the stator.

Ingredient 2: Time (The 3-Phase Supply)

If we fed direct current (DC) into these three coils, we would just get three static magnets fighting each other. Nothing would spin.

Instead, we feed our 3-phase AC system into them. Recall from Phase 0 that these currents peak sequentially, exactly 120° **apart in time**.

$$i_R(t) = I_m \sin(\omega t) \quad (1)$$

$$i_Y(t) = I_m \sin(\omega t - 120^\circ) \quad (2)$$

$$i_B(t) = I_m \sin(\omega t - 240^\circ) = I_m \sin(\omega t + 120^\circ) \quad (3)$$

Critical Misconception Alert

Do not confuse Space and Time! The physical coils are bolted down and NEVER move. Phase Y's coil is permanently stuck at a physical angle of 120° . What changes is the *amount of current* flowing through that fixed coil at any given millisecond. The magnetic field produced by Phase Y always points along the 120° axis, but it pulses stronger and weaker over time.

Block 2: The Vector Rules

Before we run the frame-by-frame animation, we must establish the rules of how these pulsating magnetic fields add together.

1. The magnetic flux (Φ) produced by a coil is directly proportional to the current (i)

flowing through it.

$$\Phi_R = \Phi_m \sin(\omega t) \quad (\text{Acts along vertical axis}) \quad (4)$$

$$\Phi_Y = \Phi_m \sin(\omega t - 120^\circ) \quad (\text{Acts along } 120^\circ \text{ axis}) \quad (5)$$

$$\Phi_B = \Phi_m \sin(\omega t - 240^\circ) \quad (\text{Acts along } 240^\circ \text{ axis}) \quad (6)$$

where Φ_m is the maximum peak flux any single phase can produce.

2. If the current is positive, the flux vector points from the un-primed letter to the primed letter (e.g., $R \rightarrow R'$). 3. If the current is negative, the flux vector points in the exact opposite direction (e.g., $R' \rightarrow R$). 4. The total magnetic field in the air gap (Φ_{res}) is the **vector sum** of these three individual fluxes at any given instant.

Block 3: Frame-by-Frame Animation (The Textual Video)

Let us slow down time. We will freeze the universe at 6 specific instants during one complete AC cycle (360°). We will calculate the strength of each magnet, draw their vectors, and find the resultant field.

Frame 1: Time $\omega t = 0^\circ$ **1. Calculate Instantaneous Currents:**

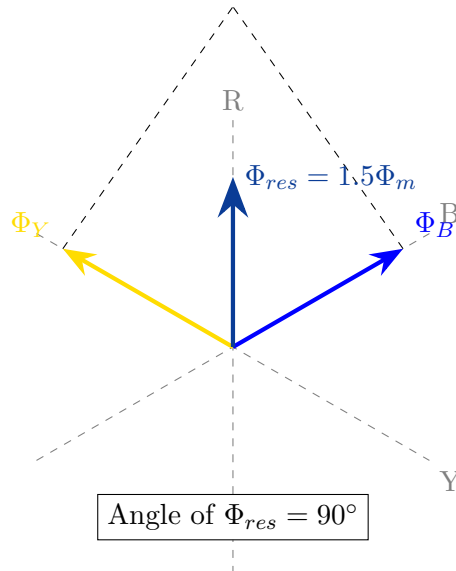
$$i_R = I_m \sin(0^\circ) = 0$$

$$i_Y = I_m \sin(0^\circ - 120^\circ) = -I_m \sin(120^\circ) = -\frac{\sqrt{3}}{2} I_m = -0.866 I_m$$

$$i_B = I_m \sin(0^\circ - 240^\circ) = I_m \sin(120^\circ) = +\frac{\sqrt{3}}{2} I_m = +0.866 I_m$$

2. Determine Flux Vectors:

- $\Phi_R = 0$. The red coil is completely dead.
- Φ_Y is highly negative (86.6% of max). It points in the reverse direction of its axis.
- Φ_B is highly positive (86.6% of max). It points in the forward direction of its axis.

3. Vector Addition:

Conclusion Frame 1: The total magnetic field is pointing straight UP, with a magnitude of 1.5 times a single coil's max strength.

Frame 2: Time $\omega t = 30^\circ$ **1. Calculate Instantaneous Currents:**

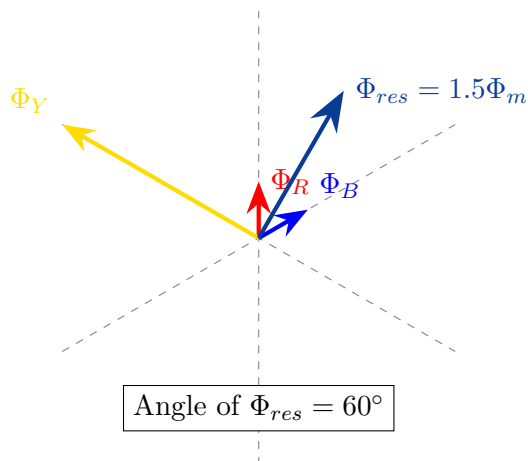
$$i_R = I_m \sin(30^\circ) = +0.5\mathbf{I}_m$$

$$i_Y = I_m \sin(30^\circ - 120^\circ) = I_m \sin(-90^\circ) = -1.0\mathbf{I}_m$$

$$i_B = I_m \sin(30^\circ - 240^\circ) = I_m \sin(-210^\circ) = +0.5\mathbf{I}_m$$

2. Determine Flux Vectors:

- Φ_R is positive (50% of max). Points UP.
- Φ_Y is at its absolute maximum negative peak (100%). Points backward along Y axis.
- Φ_B is positive (50% of max). Points forward along B axis.

3. Vector Addition:

Conclusion Frame 2: The total magnetic field is STILL exactly 1.5 in magnitude, but it has smoothly rotated 30° clockwise.

Frame 3: Time $\omega t = 60^\circ$ **1. Calculate Instantaneous Currents:**

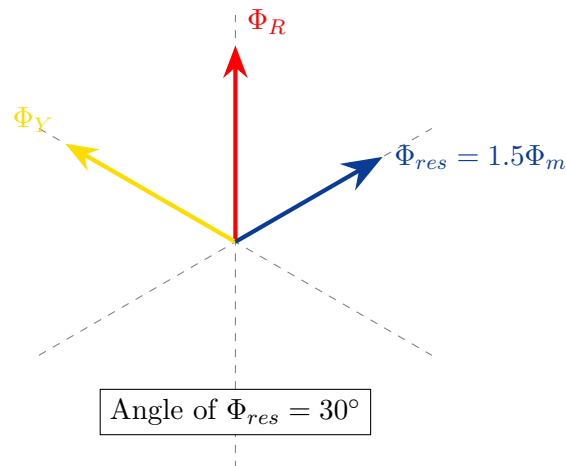
$$i_R = I_m \sin(60^\circ) = +\mathbf{0.866I_m}$$

$$i_Y = I_m \sin(60^\circ - 120^\circ) = I_m \sin(-60^\circ) = -\mathbf{0.866I_m}$$

$$i_B = I_m \sin(60^\circ - 240^\circ) = I_m \sin(-180^\circ) = \mathbf{0}$$

2. Determine Flux Vectors:

- Φ_R is highly positive (86.6% of max).
- Φ_Y is highly negative (86.6% of max).
- Φ_B is dead.

3. Vector Addition:

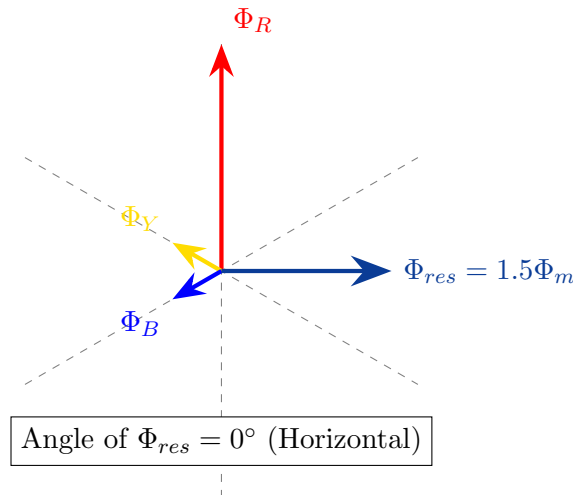
Conclusion Frame 3: The magnitude is STILL strictly 1.5. The vector has rotated another 30° clockwise.

Frame 4: Time $\omega t = 90^\circ$ **1. Calculate Instantaneous Currents:**

$$i_R = I_m \sin(90^\circ) = +1.0\mathbf{I_m}$$

$$i_Y = I_m \sin(90^\circ - 120^\circ) = I_m \sin(-30^\circ) = -0.5\mathbf{I_m}$$

$$i_B = I_m \sin(90^\circ - 240^\circ) = I_m \sin(-150^\circ) = -0.5\mathbf{I_m}$$

3. Vector Addition:

Conclusion Frame 4: We have advanced 90° in electrical time, and our magnetic field has rotated exactly 90° clockwise in physical space!

The Neurological Link (Why does this matter?)

The Pattern is Locked. We do not need to draw the remaining 320 frames. The pattern is undeniable. 1. The resulting magnetic field **never changes its size**. It is permanently stuck at $1.5 \times \Phi_m$. 2. It rotates at a perfectly constant angular velocity. We have successfully created a spinning magnet without moving a single piece of iron. But drawing frames is tedious. Is there an elegant, undeniable mathematical proof that covers all time t to back up our drawings? Yes. Let's do the rigorous math.

Block 4: The Rigorous Mathematical Proof

This proof is heavily requested in university examinations (e.g., JU 2022 Q2a, 2024 Q2a, 2025 Q2a). You must memorize the logical flow.

Rigorous Mathematical Derivation

Step 1: Define Spatial MMFs Let θ be the spatial angle measured from the axis of Phase R. The spatial distribution of MMF created by each coil is a cosine wave along its axis. Multiplying the spatial distribution by the time-varying current gives the instantaneous MMF for each phase:

$$F_R(\theta, t) = [F_m \sin(\omega t)] \cos(\theta) \quad (7)$$

$$F_Y(\theta, t) = [F_m \sin(\omega t - 120^\circ)] \cos(\theta - 120^\circ) \quad (8)$$

$$F_B(\theta, t) = [F_m \sin(\omega t - 240^\circ)] \cos(\theta - 240^\circ) \quad (9)$$

Step 2: Apply Trigonometric Expansion We use the product-to-sum trigonometric identity: $\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$

Expanding Phase R:

$$F_R = \frac{F_m}{2} [\sin(\omega t - \theta) + \sin(\omega t + \theta)] \quad (10)$$

Expanding Phase Y (Note: $A = \omega t - 120^\circ$, $B = \theta - 120^\circ$):

$$A - B = (\omega t - 120^\circ) - (\theta - 120^\circ) = \omega t - \theta \quad (11)$$

$$A + B = (\omega t - 120^\circ) + (\theta - 120^\circ) = \omega t + \theta - 240^\circ \quad (12)$$

$$F_Y = \frac{F_m}{2} [\sin(\omega t - \theta) + \sin(\omega t + \theta - 240^\circ)] \quad (13)$$

Expanding Phase B (Note: $A = \omega t - 240^\circ$, $B = \theta - 240^\circ$):

$$A - B = \omega t - \theta \quad (14)$$

$$A + B = \omega t + \theta - 480^\circ \equiv \omega t + \theta - 120^\circ \quad (\text{since } 480^\circ - 360^\circ = 120^\circ) \quad (15)$$

$$F_B = \frac{F_m}{2} [\sin(\omega t - \theta) + \sin(\omega t + \theta - 120^\circ)] \quad (16)$$

Step 3: Total Resultant Field Sum the three equations: $F_{res} = F_R + F_Y + F_B$. Group the like terms together.

$$\begin{aligned} F_{res} &= \frac{F_m}{2} [\sin(\omega t - \theta) + \sin(\omega t - \theta) + \sin(\omega t - \theta)] \\ &\quad + \frac{F_m}{2} [\sin(\omega t + \theta) + \sin(\omega t + \theta - 120^\circ) + \sin(\omega t + \theta - 240^\circ)] \end{aligned} \quad (17)$$

Step 4: The Cancellation Examine the second bracket. It contains $\sin(X)$, $\sin(X - 120^\circ)$, and $\sin(X - 240^\circ)$ where $X = \omega t + \theta$. These represent a balanced set of three three-phase vectors. **The algebraic sum of three equal vectors displaced by 120° is exactly ZERO.** The entire second bracket vanishes.

Final Result:

$$F_{res}(\theta, t) = \frac{3}{2} F_m \sin(\omega t - \theta) \quad (18)$$

Interpreting the Final Equation

The equation $F_{res} = 1.5F_m \sin(\omega t - \theta)$ is beautiful. 1. **Constant Amplitude:** The maximum value of this sine wave is exactly $1.5F_m$. It has no time dependency. It does not pulse. 2. **Traveling Wave:** The argument $(\omega t - \theta)$ represents a wave traveling in the forward direction. To track the peak of the magnetic field, we set the argument to a constant (e.g., 0):

$$\omega t - \theta = 0 \implies \theta = \omega t \quad (19)$$

Taking the derivative with respect to time gives the angular velocity of the magnetic field:

$$\frac{d\theta}{dt} = \omega \quad (20)$$

The magnetic field rotates through space exactly at the electrical angular frequency ω .

Block 5: Controlling the Beast (Direction and Speed)

Now that we have a spinning tornado of magnetic force, we need to know how to control it. How do we make the motor spin in reverse? How do we make it spin faster or slower?

5.1 Reversing the Direction of Rotation

The Neurological Link (Why does this matter?)

If you want to reverse your car, you put it in a different gear. In an induction motor, there are no gears. The rotor blindly chases the Rotating Magnetic Field. Therefore, to reverse the motor, we must reverse the direction of the RMF.

How do we reverse the RMF? We simply **swap any two of the three supply lines** connecting the grid to the motor terminals. Instead of a phase sequence of R-Y-B, we change it to R-B-Y.

Let's look at the math. If we swap Y and B, the input currents become:

$$i_R(t) = I_m \sin(\omega t) \quad (21)$$

$$i_Y(t) = I_m \sin(\omega t - 240^\circ) \quad (\text{This was B's current}) \quad (22)$$

$$i_B(t) = I_m \sin(\omega t - 120^\circ) \quad (\text{This was Y's current}) \quad (23)$$

If you run these new currents through the exact same mathematical proof from Block 4, the $(\omega t - \theta)$ terms will now cancel out to zero, and the $(\omega t + \theta)$ terms will survive and add up!

The new resultant field equation becomes:

$$F_{res, reversed} = \frac{3}{2} F_m \sin(\omega t + \theta) \quad (24)$$

Because the sign inside the sine function changed from minus to plus, the wave is now traveling in the **backward direction** ($d\theta/dt = -\omega$). The motor instantly slams into reverse.

5.2 The Speed of the RMF (Synchronous Speed)

In our derivations so far, one complete electrical cycle (360° electrical) resulted in the magnetic field completing one full physical revolution (360° mechanical) around the stator. This is true *only* if the stator is wound with a single pair of poles (2 poles: one North, one South).

What if we wind the stator with 4 poles? Or 6 poles? For a machine with P poles, one physical revolution around the machine covers $P/2$ electrical cycles.

$$\theta_{electrical} = \frac{P}{2} \theta_{mechanical} \quad (25)$$

The frequency of our electrical grid (f) is in Hertz (cycles per second). To find the mechanical speed of the magnetic field in Revolutions Per Minute (RPM), which we call the **Synchronous Speed** (N_s), we use the fundamental machine equation:

$$f = \frac{P \times N_s}{120} \implies \boxed{N_s = \frac{120f}{P}} \quad (26)$$

- If grid frequency $f = 50\text{Hz}$ and $P = 2$: $N_s = 3000$ RPM.
- If grid frequency $f = 50\text{Hz}$ and $P = 4$: $N_s = 1500$ RPM.
- If grid frequency $f = 50\text{Hz}$ and $P = 6$: $N_s = 1000$ RPM.

By choosing how we wind the coils inside the stator (the number of poles P), we can permanently set the baseline speed of the motor.

Phase 1 Completion Checklist

You have successfully constructed the Rotating Magnetic Field in your mind and proven it mathematically. This is the hardest conceptual hurdle in Electrical Machines.

Ensure you can answer "Yes" to these checkpoints before moving to Phase 2:

- [] Do I understand that the stator coils do not physically move, only the magnetic intensity they project changes?
- [] Can I reproduce the trigonometric proof that proves $F_{res} = 1.5F_m$?
- [] If someone asks me how to reverse an induction motor, can I instantly answer "swap any two supply lines" and explain the math behind it?
- [] Can I calculate the Synchronous Speed (N_s) if given the grid frequency and pole count?

The Neurological Link (Why does this matter?)

The Bridge to Phase 2: We now have a massive, invisible magnetic field spinning at 1500 RPM inside our stator. Inside this spinning field, we have placed our heavy iron rotor wrapped in short-circuited copper bars.

According to Faraday and Lenz (from Phase 0), what will the rotor do? It will try to chase the spinning field. But will it ever catch it? What happens if it *does* catch it?

In Phase 2, we will explore the kinematics of the rotor, introduce the concept of "Slip", and finally prove why the Induction Motor is nothing more than a Transformer that learned how to spin.